

Vol 11 Chapter 9 — Cyclotomic Fields and Reed-Solomon

Keeper (author pass — deep math/physics revision)

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Chapter 9 — Cyclotomic Fields and Reed-Solomon

Legacy chapter (May 2026). Last accuracy-checked 2026-09-11 (Keeper, K1892): the presentation scan found no retired reading here, and the tiers in the front matter are the May tiers, not re-verified line by line. Where this chapter and the Spine (Curriculum/Spine_DIV5_QM_GR_SM/) disagree, the Spine is current; where the Spine and the theorem registry disagree, the registry wins.

Level 1 — one sentence

Cyclotomic fields $\mathbb{Q}(\zeta_N)$ — obtained by adjoining primitive N th roots of unity — are workhorse algebraic structures, and the BST substrate uses Reed-Solomon coding on the finite field $\text{GF}(2^8) = \text{GF}(128)$ for its native information-channel operations (K59 cyclotomic mechanism RATIFIED, Vol 14 Ch 2).

Level 2 — graduate-physicist precision

9.1 Cyclotomic fields

$\mathbb{Q}(\zeta_N)$ where $\zeta_N = e^{2\pi i/N}$ a primitive N th root of unity.

Degree over $\mathbb{Q}$: $\varphi(N)$ (Euler totient).

Galois group: $(\mathbb{Z}/N\mathbb{Z})^\times$, abelian.

Kronecker-Weber: every abelian extension of $\mathbb{Q}$ is contained in some $\mathbb{Q}(\zeta_N)$.

9.2 Finite fields

For prime p and integer $n \geq 1$: unique finite field $\text{GF}(p^n) = \mathbb{F}_{p^n}$ of p^n elements.

For BST: $\text{GF}(2^7) = \text{GF}(128)$ — characteristic 2, dimension 7 over $\mathbb{F}_2$.

The 7 = BST primary g. Substrate uses $\text{GF}(128)$ as natural symbol alphabet.

9.3 Reed-Solomon codes

Reed-Solomon 1960: error-correcting code based on polynomial evaluation over $\text{GF}(q)$.

For $\text{RS}(n, k)$ code over $\text{GF}(q)$ with $n \leq q$: - Encode: message $m \in \text{GF}(q)^k \rightarrow$ polynomial $p_m(x)$ of degree $< k \rightarrow$ codeword $(p_m(\alpha_1), \dots, p_m(\alpha_n))$ at evaluation points $\alpha_i \in \text{GF}(q)$ - Minimum distance: $d = n - k + 1$ (MDS — maximum distance separable) - Corrects up to $\lfloor (d - 1)/2 \rfloor$ errors

Industrial uses: CD/DVD storage, deep-space communication (Voyager), QR codes, RAID storage.

9.4 BST substrate RS on $\text{GF}(128)$

K59 (Spring 2026 RATIFIED): cyclotomic mechanism framework. Substrate uses Reed-Solomon on $\text{GF}(128)$ for its natural information-channel operations.

Per-Koons-tick: substrate emits one RS-coded symbol from $\text{GF}(128)$. Symbol rate $\sim 1/t_K \sim 10^{120}$ symbols/sec per substrate K-type degree of freedom.

The 7-step cyclotomic cascade RG (Vol 6 Ch 9, K59) is the substrate's natural RG flow through 7 cyclotomic steps.

9.5 Why $\text{GF}(2^g)$ specifically

BST argues $\text{GF}(2^g) = \text{GF}(128)$ is substrate's unique natural choice: - $g = 7$ is the BST primary - g prime (so $\text{GF}(2^g)$ is irreducible — only sub-field is $\text{GF}(2)$) - $2^g = 128 = 137 - g - 2$ ($N_{\max}$ -related) - Maximum RS codeword length $n = 127 = 2^g - 1$ — Mersenne!

9.6 K-audit anchors

- K59 RATIFIED: cyclotomic mechanism framework
- Paper #122: Information Substrate (substrate as RS-coded channel)
- Vol 14 Ch 2: Reed-Solomon on $\text{GF}(128)$ in info-theory volume

Level 3 — 5th-grader accessibility

Cyclotomic fields $\mathbb{Q}(\zeta_N)$ = rationals with primitive N th root of unity added. Finite fields $\text{GF}(p^n)$ are the algebraic playgrounds for coding theory. Reed-Solomon codes ($n = q$ symbols from $\text{GF}(q)$, k message) are used in CDs, DVDs, satellite communication, deep-space probes. BST substrate uses Reed-Solomon coding on $\text{GF}(128) = \text{GF}(2^7) = \text{GF}(2^g)$ — the 7 is BST primary g . Substrate emits RS-coded symbols at the Koons-tick rate. K59 ratified the cyclotomic mechanism framework.

What comes next

Chapter 10 develops Mersenne primes.

Where to look this up

- Washington, *Introduction to Cyclotomic Fields*
- MacWilliams-Sloane, *The Theory of Error-Correcting Codes*
- BST: K59; Paper #122; Vol 14 Ch 2